PRD: TimeTalk, Conversational Timesheet Copilot

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TimeTalk is a concept prototype for a time-tracking product: employees log time by chatting, the copilot fills the timesheet, flags anomalies, and a live AI Value panel proves its ROI per conversation. All data is mock; no real company branding.

Problem statement

In services firms, every hour logged feeds billing, project costing and payroll, yet employees treat timesheets as a chore. They fill them late, from memory, by clicking through project and task dropdowns, so entries are missing, vague or charged to the wrong project.

Bad time data leaks straight into revenue: unbilled hours, invoice disputes and late month-end close. Adding a chatbot alone does not fix this; leaders will only trust AI in billing data if it is accurate, explains itself, never submits without the user, and proves it is worth its cost.

Goals and KPIs

The prototype must show two things in one demo: logging time by chat is faster and more accurate than the form, and the AI proves its own value.

Goal	KPI	Demo target
Faster entry	Time to log a day, chat vs manual form	Under 30 seconds vs ~4 minutes
Accurate entries	Entries confirmed without edits	90%+ on the demo scripts
Safe automation	Entries submitted without user confirmation	0
Catch billing leakage	Anomalies flagged before submit	Every seeded anomaly caught
Prove value	Value-to-cost ratio per conversation	Shown live, above 10x

Goal	KPI	Demo target
Transparency	Answers showing reasoning and confidence	100%

Targets are design goals for the demo, not measured results.

Personas

Persona	Pain today	What TimeTalk gives them
Priya, consultant (primary)	Logs time Friday evening from memory, hunting for project codes	Types one sentence; entries appear pre-filled for one-click confirm
Arjun, project manager	Chases missing or misfiled hours before invoicing	Anomaly flags before submit, so fewer corrections later
Finance / billing lead	Unbilled hours and disputes at month-end	Cleaner billable data and fewer write-offs
AI program owner	Must justify AI spend to leadership	AI Value panel with cost, time saved and ROI per conversation

User flow

1. **Describe:** Priya types "Spent 3 hours on the Acme design review yesterday and 1 hour on internal standup."
2. **Parse:** the LLM extracts date, hours, project, task and billable flag for each activity as structured JSON.
3. **Match:** each activity is matched to a project and task from the mock catalogue, with a confidence score.
    - Confidence 80% or above: shown as a draft row.
    - Below 80%, or more than one likely match: TimeTalk asks a short question ("Acme Website or Acme Mobile?") instead of guessing.
4. **Check:** anomaly rules run on the draft rows and attach warnings (for example, billable time on an internal project).

5. **Explain:** each row shows why it was filled that way ("Matched 'Acme design review' to ACME-WEB / Design, 92%").
6. **Confirm:** Priya edits if needed and clicks Confirm; nothing is written to the timesheet before this.
7. **Measure:** the AI Value panel updates with tokens, cost, time saved and running ROI for the conversation.

Functional requirements

ID	Requirement	Priority
FR1	Chat box accepts free-text descriptions of one or more activities, including relative dates ("yesterday", "Monday")	Must
FR2	LLM returns structured JSON per activity: date, hours, project, task, billable, confidence, reason	Must
FR3	Match project and task only from the mock catalogue; never invent a project code	Must
FR4	Ask a clarifying question when confidence is below 80% or matches are ambiguous	Must
FR5	Show draft rows in a weekly timesheet table, editable before confirm	Must
FR6	Run anomaly checks on drafts and show warnings inline	Must
FR7	Write rows to the timesheet only after the user clicks Confirm	Must
FR8	AI Value panel updates after every message	Must
FR9	"Why?" toggle on each row reveals the reasoning trace	Should
FR10	Admin view: totals across all demo conversations (cost, time saved, ROI, edit rate)	Should
FR11	Voice input	Won't (this version)

Anomaly checks

Anomalies are caught by deterministic rules and a simple statistical check, not the LLM, so they are predictable, cheap and explainable. This is the "traditional ML alongside GenAI" layer.

Check	Logic	Warning shown
Unusual daily hours	Day total more than 2 standard deviations above the user's 8-week average (z-score)	"11 hrs is well above your usual 8.2 hrs. Correct?"
Weekend or holiday	Entry dated Saturday, Sunday or a mock holiday	"This is a Sunday. Still billable?"
Billable on internal	Billable flag on a project marked internal	"Internal projects aren't billable. Switch to non-billable?"
Budget burn	Hours push a task past 90% of its budget	"ACME-WEB Design is at 94% of budget."
Closed project	Entry against a project past its end date	"This project closed on 31 Oct."
Duplicate	Same project, task and date already logged	"You already logged 3 hrs here today."

AI Value panel

The panel answers one question live: is this AI worth what it costs? It sits beside the chat and updates after every message.

Metric	Definition
Tokens	Input + output tokens for the conversation, from the LLM API response
AI cost (₹)	Tokens × model price per token (set in config)
Time saved (min)	Manual baseline per entry (configurable, default 2 min) × confirmed entries, minus time spent in chat

Metric	Definition
Value of time saved (₹)	Time saved × loaded hourly cost (configurable, default ₹1,500/hr)
Billing leakage caught (₹)	Billable hours recovered by anomaly fixes × mock bill rate
Edit rate	Confirmed rows the user edited ÷ all confirmed rows (quality signal)
ROI	Total value ÷ AI cost

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\text{ROI} = \frac{\text{Value of time saved} + \text{Billing leakage caught}}{\text{AI cost}}
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All defaults are demo assumptions shown in a small "Assumptions" footer, so viewers can see exactly how ROI is calculated.

AI quality, observability and controls

TimeTalk drafts, the user decides: the AI never writes billing data on its own.

Area	Control
Grounding	LLM receives only the mock project and task catalogue; output validated against it, unknown codes rejected
Structured output	JSON schema enforced; malformed output retried once, then the user is asked to rephrase
Confidence	Below 80% or ambiguous match triggers a clarifying question, never a guess
Human in the loop	Draft rows only; Confirm is the single path to the timesheet
Observability	Each row's reasoning, confidence, model, tokens and latency are logged and visible via "Why?"
Privacy	Only the chat text and catalogue go to the LLM; no personal or payroll data

Area	Control
Cost guardrail	Max tokens per message capped; cost per conversation shown live
Progressive autonomy	Future idea: auto-confirm high-confidence recurring entries (like daily standup) once a user's edit rate stays low

Mock data spec

One fictional services firm, "Northwind Consulting", one user (Priya) and these projects, stored as a JSON file:

Code	Client / project	Tasks	Type	Budget used	Bill rate (₹/hr)
ACME-WEB	Acme Website Redesign	Design, Development, QA	Billable	88%	4,000
ACME-MOB	Acme Mobile App	Design, Development	Billable	40%	4,000
GLOBEX-DATA	Globex Data Migration	Analysis, Migration	Billable	65%	3,500
INITECH-AUD	Initech Audit Support	Fieldwork	Billable, closed 31 Oct	100%	3,000
INT-OPS	Internal	Standup, Training, Admin	Non-billable	n/a	0

Also seed 8 weeks of Priya's past entries (about 8.2 hrs/day average) so the z-score check has a baseline, and one mock holiday.

Build plan, demo and scope

Stack: React + Vite front end, FastAPI back end, Groq LLaMA-3.3 or Gemini Flash with JSON output, mock data in JSON files, deployed on Railway. Same stack as Groww Pulse.

Phases in Antigravity (2 to 3 days):

1. **Day 1, core loop:** mock data files, chat box, LLM parsing to JSON, catalogue matching with confidence, draft rows and Confirm.

2. **Day 2, trust layer:** clarifying questions below 80%, anomaly rules with inline warnings, "Why?" reasoning toggle.
3. **Day 3, value and polish:** AI Value panel with assumptions footer, deploy to Railway, record the Loom.

90-second demo script:

1. "Spent 3 hours on the Acme design review yesterday and 1 hour on standup." Two correct rows, one billable and one internal. (Speed.)
2. "2 hours on Acme development." TimeTalk asks: Website or Mobile? (Never guesses.)
3. "4 hours on Initech fieldwork today, billable." Flags the closed project. (Leakage caught.)
4. Open "Why?" on a row. (Observability.)
5. End on the AI Value panel: cost in paise, minutes saved, ROI. (Cost vs value.)

Out of scope: real authentication, integration with any real time-tracking product, approvals workflow, payroll, voice, mobile app.

Branding: fictional name and mock data only; no real company names, logos or UI copied.